**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 156 · Timed Paper 1 Practice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the purpose of the Data Protection Act 2018?

Q2. Why indent code?
A. To fix every logic error
B. To replace all conditions
C. To change false into true
D. To show blocks clearly

Q3. “A comment always changes the value of the variable on the next line.” Correct this. Explain why.

Q4. An indicator uses Q = NOT A. Give outputs for A=0 and A=1, then evaluate NOT(NOT A) for both values.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Paper 1 timing — The OCR J277/01 paper has 80 marks in 90 minutes.

Show calculations — Write formula, substitute numbers and give final units.

Plan explanations — Match distinct points and justifications to mark tariff.

Track time — Leave opportunity to revisit unanswered questions without sacrificing required detail.

## New learning

Paper 1 timing: The OCR J277/01 paper has 80 marks in 90 minutes.

Show calculations: Write formula, substitute numbers and give final units.

Plan explanations: Match distinct points and justifications to mark tariff.

Track time: Leave opportunity to revisit unanswered questions without sacrificing required detail.

## Worked example

Model a four-mark image-size calculation: 80×50 pixels at 8 bits/pixel.

1. Count 80×50=4000 pixels.
2. Multiply by eight: 32,000 bits.
3. Divide by eight: 4000 bytes.
4. Check that metadata/compression are excluded.

# We Do · Practise with guidance

### Guided 1

Plan a six-mark comparison of SSD and HDD for a traveller.

Worked method: Balance with higher cost per GB; link to travel needs.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Convert a raw data size from bits to bytes.

```
sizeBits = 24000
sizeBytes = sizeBits DIV 8
print(sizeBytes)
```

Predict the byte count.

Change sizeBits to 48000.

Calculate the original result in decimal kB using 1000 bytes per kB.

### Question 1

State RAM volatility and its implication. [2 marks]

### Question 2

Calculate the largest eight-bit unsigned number. [2 marks]

### Question 3

Explain why a router is needed between separate networks. [3 marks]

## Exit Ticket · Answer before checking

Units can be ignored as long as the final calculation looks plausible.







## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It sets rules for using personal data lawfully and protects people's rights.

Q2. D — To show blocks clearly

Q3. It explains important intent or reasoning without misleading the reader.

Q4. (a) Q: 1, 0. Double NOT: 0, 1, restoring A.

### I Do / We Do pages · Guided practice

Q1. Balance with higher cost per GB; link to travel needs.

### You Do pages · Code First

Expected result: 24,000 bits ÷ 8 = 3,000 bytes (3 kB using decimal units). Changing to 48,000 bits gives 6,000 bytes (6 kB). This is a unit calculation, not a complete file-size formula.

### You Do pages · Independent questions

Q1. Volatile; active contents are lost without power.

Q2. 255.

Q3. Forwards packets using logical/IP routing, allowing communication between networks.

### Exit Ticket · Model answer

False

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**